

Changyang Li

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Education

George Mason University

PhD in Computer Science
Advisor: Lap-Fai (Craig) Yu

Aug 2019 - June 2024

University of Virginia

MS in Computer Science

Aug 2017 - May 2019

Beijing Institute of Technology

BS in Computer Science

Aug 2013 - June 2017

Experience

GoerTek Alpha Labs

Research Scientist

Scene perception, generation and reconstruction, human-scene interaction.

Feb 2025 - Current

Meta

Computer Vision Engineer

Parametric human body/hand model, human-object interaction.

June 2024 - Jan 2025

Design Computing and Extended Reality (DCXR) Group

Graduate Research Assistant

Interactive XR system, human-scene interaction, human-centered scene perception.

Aug 2019 - June 2024

Mentor: Lap-Fai (Craig) Yu

Center for Vision, Cognition, Learning, and Autonomy

Research Intern

May 2018 - Aug 2018

Mentor: Song-Chun Zhu

Publications

Crafting Dynamic Virtual Activities with Advanced Multimodal Models

Changyang Li, Qingan Yan, Minyoung Kim, Zhan Li, Yi Xu, Lap-Fai Yu

2025 IEEE International Symposium on Mixed and Augmented Reality (ISMAR), 2025

Dragon's Path: Synthesizing User-Centered Flying Creature Animation Paths for Outdoor Augmented Reality Experiences

Minyoung Kim, Rawan Alghofaili, **Changyang Li**, Lap-Fai Yu

SIGGRAPH 2024 Conference Papers

Generating Activity Snippets by Learning Human-Scene Interactions

Changyang Li, Lap-Fai Yu

ACM Transactions on Graphics (Proceeding of SIGGRAPH 2023)

Location-Aware Adaptation of Augmented Reality Narratives

Wanwan Li*, **Changyang Li***, Minyoung Kim, Haikun Huang, Lap-Fai Yu

Proceedings of the 2023 CHI Conference on Human Factors in Computing Systems

Optimizing Product Placement for Virtual Stores

Wei Liang, Luhui Wang, Xinzhe Yu, **Changyang Li**, Rawan Alghofaili, Yining Lang, Lap-Fai Yu

2023 IEEE Conference on Virtual Reality and 3D User Interfaces (VR)

Establishing Design Computing and Extended Reality Facilities for Remote Virtual Reality Training

Lap-Fai Yu, **Changyang Li**, Yongqi Zhang, Rawan Alghofaili, Haikun Huang, Liuchuan Yu, Huimin Yu, Minsoo Choi, Brenda Bannan, Christos Mousas

2023 IEEE Conference on Virtual Reality and 3D User Interfaces Abstracts and Workshops (VRW)

Interactive Augmented Reality Storytelling Guided by Scene Semantics

Changyang Li, Wanwan Li, Haikun Huang, Lap-Fai Yu

ACM Transactions on Graphics (Proceeding of SIGGRAPH 2022)

Synthesizing Scene-Aware Virtual Reality Teleport Graphs

Changyang Li, Haikun Huang, Jyh-Ming Lien, Lap-Fai Yu

ACM Transactions on Graphics (Proceeding of SIGGRAPH Asia 2021)

A review on virtual reality skill training applications

Biao Xie, Huimin Liu, Rawan Alghofaili, Yongqi Zhang, Yeling Jiang, Flavio Destri Lobo, **Changyang Li**, Wanwan Li, Haikun Huang, Mesut Akdere, Christos Mousas, Lap-Fai Yu

Frontiers in Virtual Reality. 2021

Learning Virtual Grasp with Failed Demonstrations via Bayesian Inverse Reinforcement Learning

Xu Xie*, **Changyang Li***, Chi Zhang, Yixin Zhu, Song-Chun Zhu

2019 IEEE/RSJ International Conference on Intelligent Robots and Systems (IROS) 2019

Earthquake Safety Training through Virtual Drills

Changyang Li, Wei Liang, Chris Quigley, Yibiao Zhao, Lap-Fai Yu

IEEE Transactions on Visualization and Computer Graphics (Special Issue on IEEE VR 2017)

Joint labelling and segmentation for 3D scanned human body

Hanqing Wang, **Changyang Li**, Zikai Gao, Wei Liang

SIGGRAPH ASIA 2016 Virtual Reality meets Physical Reality: Modelling and Simulating Virtual Humans and Environments

Academic Services and Activities

Program Committee

- AAAI 2026
- IEEE VR 2026
- IEEE ISMAR 2025

Conference Paper Reviewer

- SIGGRAPH 2023, 2024; SIGGRAPH Asia 2023
- CHI 2024; UIST 2023
- Pacific Graphics 2023
- IEEE VR 2018, 2019
- IROS 2024

Journal Paper Reviewer

- ACM Transactions on Graphics 2023, 2024
- IEEE Transactions on Visualization and Computer Graphics 2024
- Computer Graphics Forum 2022, 2023, 2025

Invited Talks

- IEEE Metaverse Seminar, 2022
- Autodesk 'Research Connections' Talk Series, 2024

Skills

Programming Python, C/C++/C#, HTML/CSS, Swift

Deep Learning Pytorch, Keras, Tensorflow

ML/CV/CG Numpy, Scikit-learn, SciPy, OpenCV, OpenGL, Matlab

AR/VR ARCore, ARKit, RealityKit, Oculus Quest, Microsoft HoloLens, HTC Vive

Game Engine Unity 3D, Unreal Engine